

Ultracold interactions between a single Barium ion and an atomic gas of Lithium

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BaLi: What and Why?

What do we do?

- ▶ Hybrid traps
- ▶ $^{138}\text{Ba}^+$ ions and neutral ^6Li atoms
- ▶ Investigate interactions in the ultracold regime

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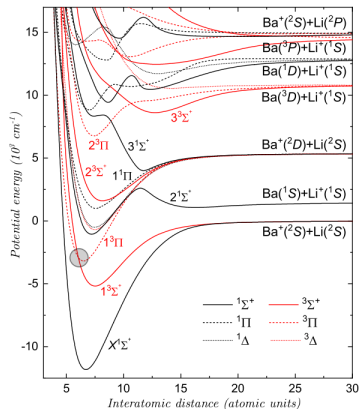
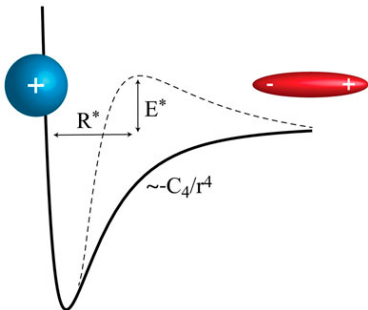
Why hybrid traps?

- ▶ Longer range interactions than neutrals (r^{-4} vs r^{-6})
- ▶ Cold quantum chemistry
- ▶ Quantum simulation

Atom-Ion interaction

The ion induces a dipole moment in the atom ($p \propto E(r) \propto \frac{1}{r^2}$)

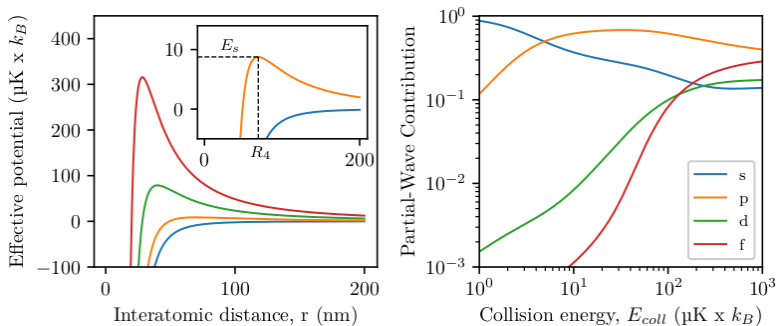
$$V_{\text{atom-ion}}(r) \approx -\frac{C_4}{r^4}$$



Atom-Ion interaction

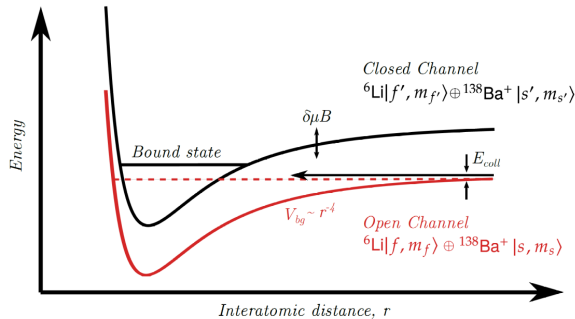
The total wavefunction is composed of partial waves of angular momentum l .
The contribution of partial waves depends on the collision energy.

$$V_{\text{atom-ion}}(r) \approx -\frac{C_4}{r^4} + \frac{\hbar^2 l(l+1)}{2\mu r^2}$$



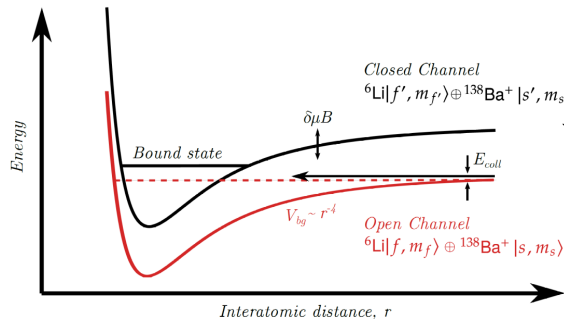
Feshbach Resonances

An incoming state (open channel) can couple to a bound state (closed channel). As the bound state approaches the incoming state the mixing between states increases.

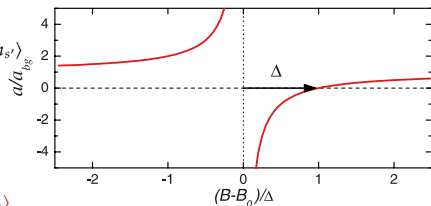


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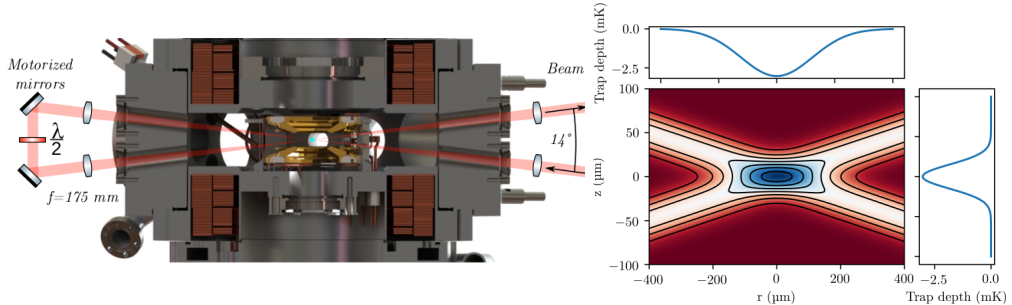
$$a = a_{bg} \left(1 - \frac{\Delta}{B - B_0} \right)$$



Chin2010

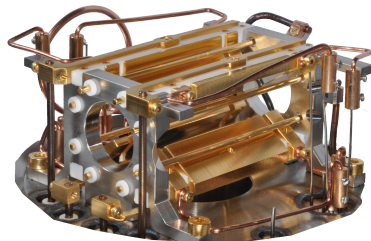
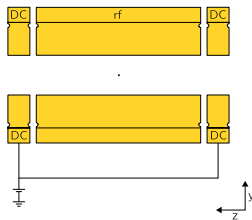
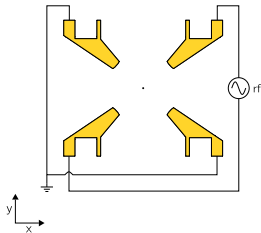
The Atom Trap

Atoms trapped in a crossed optical dipole trap (xODT).
Evaporatively cooled to $T_{\text{Li}} \approx 0.7 - 20 \mu\text{K}$ and spin polarised.
Atom number $N \sim 10^4$, density $n \sim 10^{13} \frac{1}{\text{cm}^3}$.



The Ion Trap

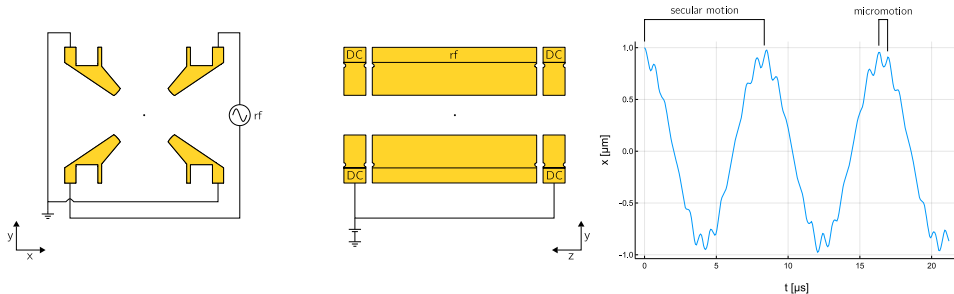
Ion confined in linear rf trap $\Omega_{\text{rf}} = 1.4 \text{ MHz}$, Doppler cooled to $T_{\text{Ba}^+} \approx 300 \mu\text{K}$



Effective potential depth $\sim 10^3 \text{ K}$ traps ion for hours

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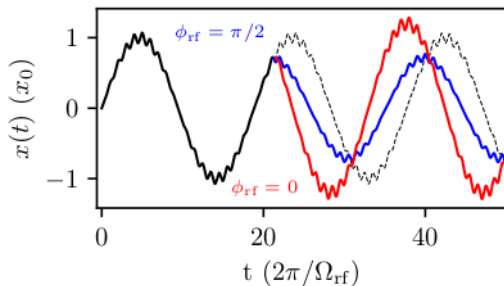
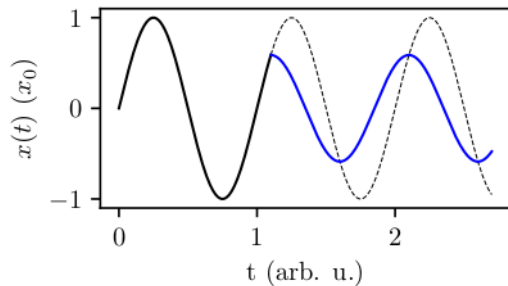


Ion's motion includes slower secular part ($\omega_r \approx 100 \text{ kHz}$) and fast micromotion part (Ω_{rf}) dc fields increase micromotion:

$$r_i(t) = [1 - q_i \cos(\Omega_{\text{rf}} t)] \cdot [x_{\text{dc}} + x_0 \cos(\omega_r t)]$$

Collisions

Sympathetic cooling - but ...



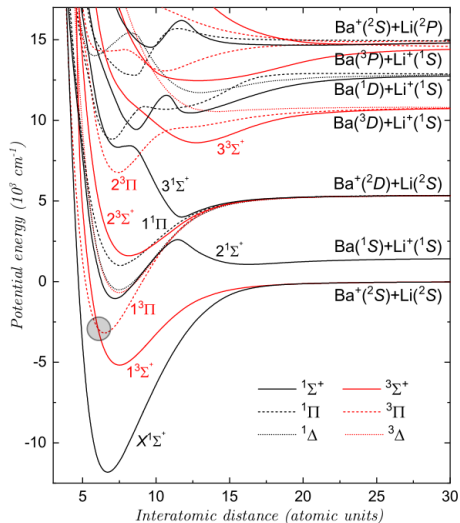
T_{ion} limited by rf trap

$^{138}\text{Ba}^+ - ^6\text{Li}$

- ▶ $^{138}\text{Ba}^+$ is hydrogen-like
- ▶ ^6Li is fermionic
- ▶ Large mass ratio
- Collision energy dominated by (colder) Li

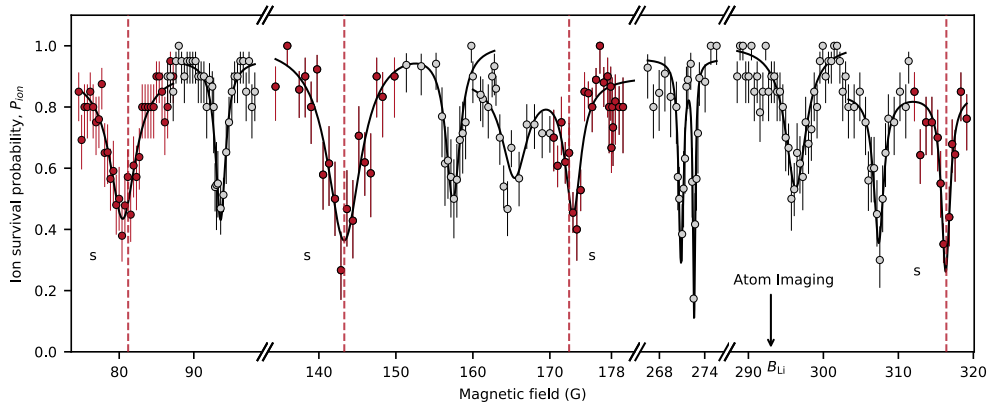
$$E_{\text{coll}} = \frac{\mu}{m_{\text{Li}}} E_{\text{Li}} + \frac{\mu}{m_{\text{Ba}^+}} (E_{\text{Ba}^+} + E_{\text{mm}})$$

- Also reduces the effect collisional heating
- ▶ Individual ground states also molecular ground state - low background losses



Feshbach Resonances: Spectrum for Li in $|2\rangle$

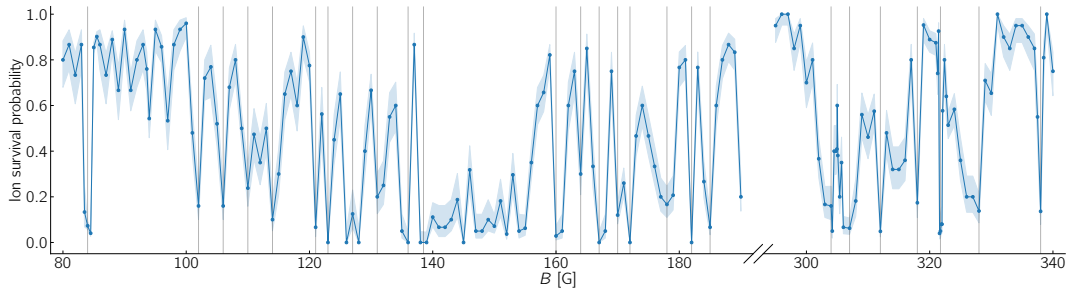
First looked at Li in $|f = 1/2, m_F = -1/2\rangle = |2\rangle$



Free fit of a_S and a_T including coupling to s , p and d assigns four s -wave resonances
(Collaboration with M. Tomza and K. Jachymski in Warsaw)

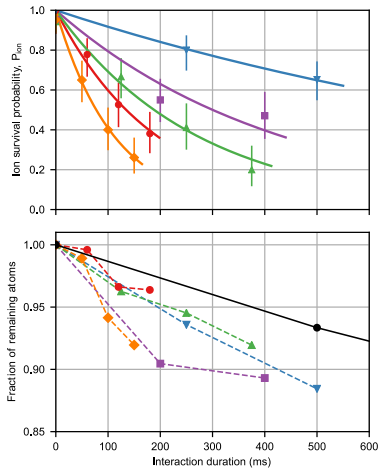
Feshbach Resonances: Spectrum for Li in $|1\rangle$

Then $|f = 1/2, m_F = 1/2\rangle = |1\rangle$



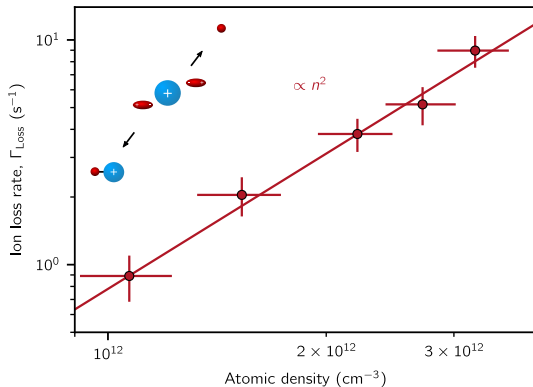
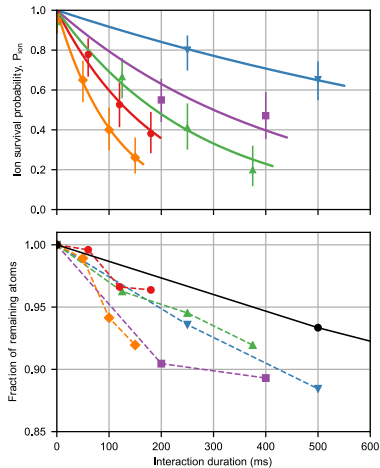
which has a much busier spectrum - more work for the theorists!

Density Dependence of Ba^+ Loss on Resonance



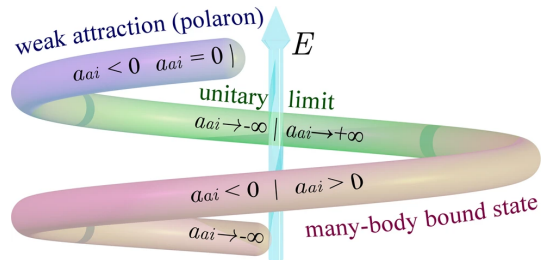
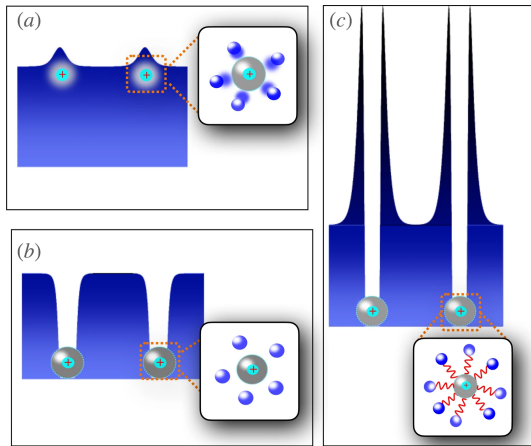
Vary atomic density and probe ion decay rate.

Density Dependence of Ba^+ Loss on Resonance



Vary atomic density and probe ion decay rate.
 $\Rightarrow$ TBR is dominant loss process on resonance!

Outlook: Many-Body Physics



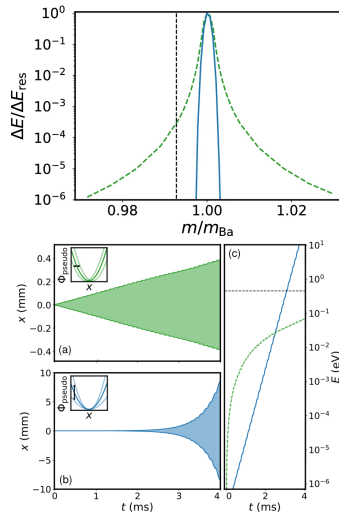
astrakharchik2021ionic

astrakharchik2023many

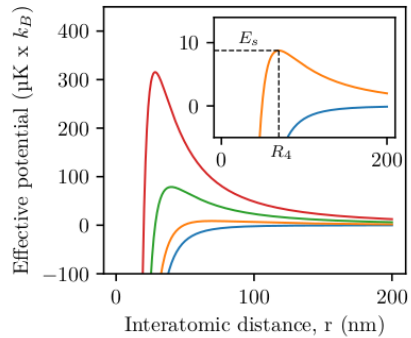
Outlook: Many-Body Physics

- ▶ Weak and strong coupling both $\rightarrow \delta m_{\text{eff}}$
 - $\rightarrow$ "'tickling"' at motional frequency to measure mass
- ▶ Weak coupling
 - ▶ Able to resolve the shift?
 - $\rightarrow$ Parametric excitation
- ▶ Strong coupling
 - ▶ Do MBBSs live long enough?

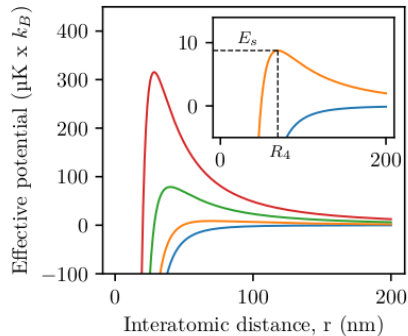
schmidt2020mass



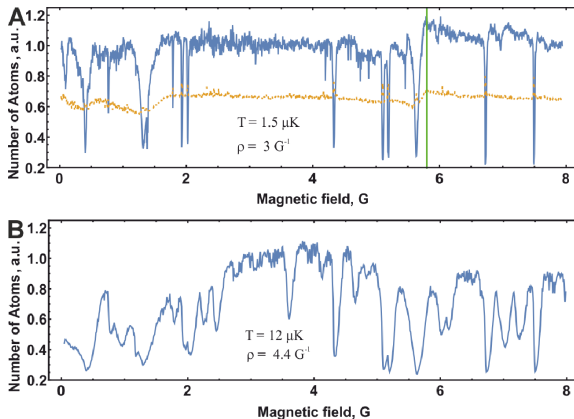
Outlook: Energy Dependence



Outlook: Energy Dependence



Experimental data in thulium:



Khlebnikov2019RandomTC

Outlook: Energy Dependence

- ▶ Increase ${}^6\text{Li}$ bath temperature
 - ▶ Evaporate to higher temp
 - more light → light assisted interaction?
 - ▶ Adiabatic ramps?
- ▶ Increase ${}^{138}\text{Ba}^+$ energy
 - ▶ Offset fields to increase micromotion
 - increase in rf power → rf association?
 - displacement → inhomogeneities?
 - ▶ Additional rf
 - increase in rf power → rf association?

Outlook: All-Optical

- ▶ Optical trapping of ions demonstrated for $t_{\text{trap}} \gg t_{\text{int}}$
- ▶ No micromotion!
- ▶ ...but even more light
 - light assisted interactions?

Thank you!



BaLi Team:
Fabian Thielemann, Joachim Welz, Wei
Wu, Pascal Weckesser (former)
Tobias Schätz

Theory collaborators:
Michał Tomza,
Krzysztof Jachymski,
Agata Wojciechowska



QUSTEC
Quantum Science and Technologies
at the European Campus



Bibliography